Project Report

visualization tool for electric vehicle charge and range analysis

Team ID : LTVIP2026TMIDS36743

Team Members:

Mannena Sasikala (Leader)

Moyya Ashajyothi

Manem Dhilleswari

1.INTRODUCTION

Project Overview

The **Visualization Tool for Electric Vehicle Charge and Range Analysis** is designed to monitor, analyze, and predict electric vehicle (EV) battery performance and driving range using real-time and historical data. As EV adoption increases, users often face challenges such as inaccurate range estimation, inefficient charging patterns, and lack of detailed battery performance insights. This project aims to address these issues through an interactive and data-driven dashboard system.

The system collects real-time data from EV sensors, including State of Charge (SoC), voltage, current, temperature, speed, and energy consumption. This data is processed using analytical models to estimate the remaining driving range and evaluate battery health. The processed information is then displayed through a user-friendly visualization dashboard that presents clear charts, graphs, and performance indicators.

The tool helps EV users and fleet operators make informed decisions regarding trip planning, charging schedules, and battery maintenance. By integrating predictive analytics with real-time monitoring, the system reduces range anxiety, improves energy efficiency, and enhances overall vehicle reliability.

Overall, this project demonstrates how data visualization and analytics can be effectively applied to improve electric vehicle performance monitoring and optimize charging and range management.

2.IDEATION PHASE

1.Problem Statement:

Analysing different data from Multiple sources for Electric cars in India and Globally. We have 4 Different datasets we need to analyse the data and create Dashboard and story that can represent the data and show the Visuals for the data.

With the rapid growth of electric vehicles (EVs) in cities like Vijayawada and across India, users face challenges in understanding battery performance, charging patterns, and driving range. Many EV owners experience range anxiety, lack proper insights into battery consumption, and cannot easily analyze how factors such as speed, temperature, charging time, and driving conditions affect vehicStatement

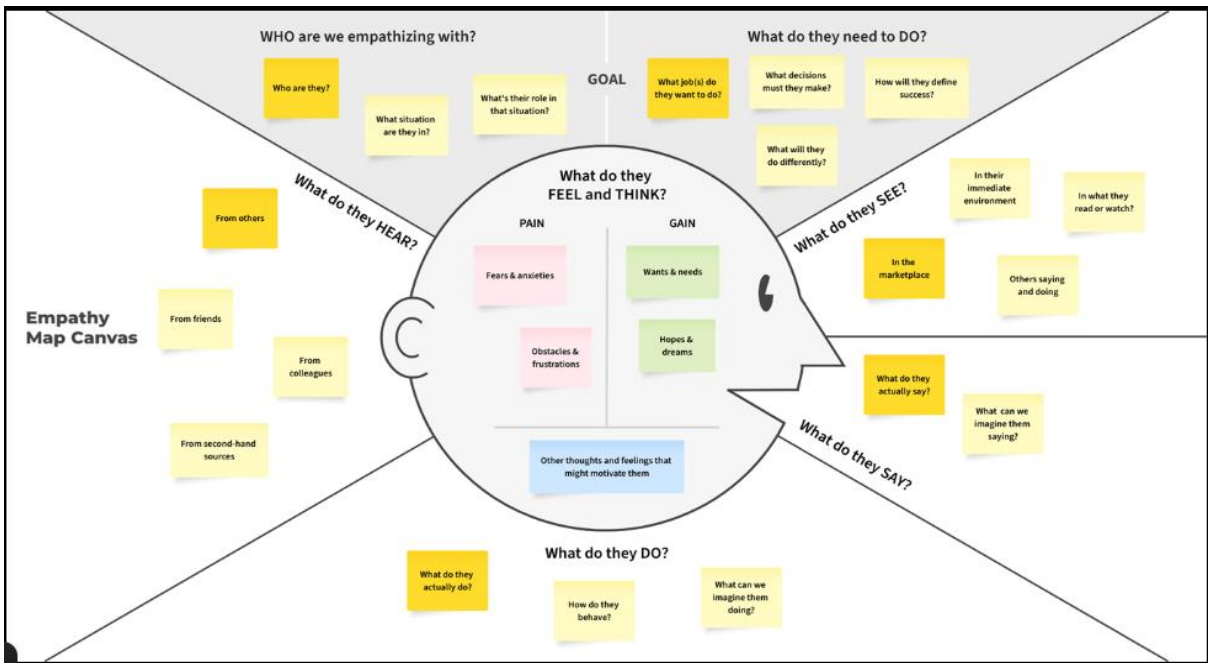
The increasing adoption of Electric Vehicles (EVs) has created a need for better monitoring and analysis of battery charging and driving range. Most existing EV systems only display basic details such as battery percentage and estimated range, without providing detailed insights into battery performance, energy consumption, and charging patterns.

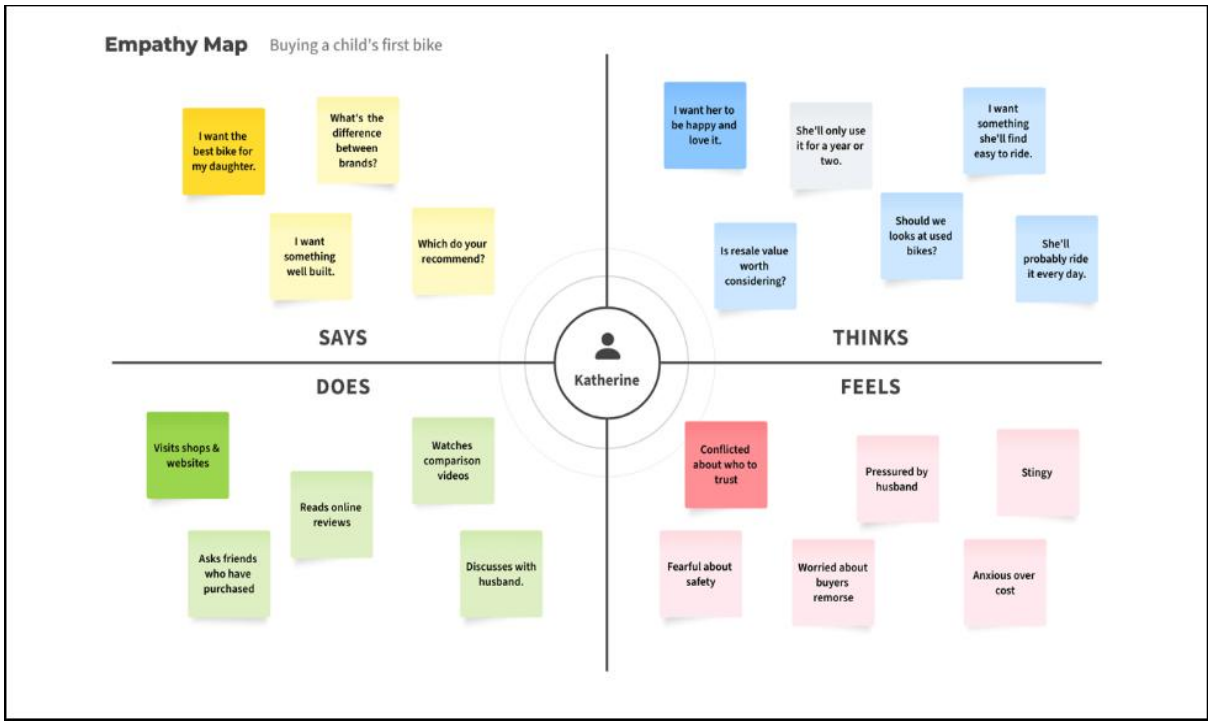
Due to the lack of proper data visualization and analytical tools, EV users often face range anxiety, inefficient trip planning, and difficulty understanding the factors that affect battery efficiency. Fleet operators and researchers also lack a centralized system to analyze EV performance data effectively.

Therefore, there is a need to develop a Visualization Tool for Electric Vehicle Charge and Range Analysis that collects, analyzes, and presents EV data (such as battery level, charging time, energy usage, distance traveled, and speed) through interactive graphs and dashboards.

This system will help users better understand charging behavior, predict driving range more accurately, improve battery management, and support informed decision-making for sustainable transportation.

2.Empathy Map Canvas





The Empathy Map Canvas is a strategic tool used to gain a deep understanding of a user's experiences, emotions, and behaviors. At its core, it explores what the user thinks and feels, focusing on their challenges and aspirations. Surrounding this central insight, the map examines the user's identity, goals, and the tasks they need to accomplish, alongside the environment they see and the influences they hear from their social and informational circles. It also considers what users say aloud or might say, as well as their actual behaviors. By addressing these aspects, the empathy map creates a comprehensive, human-focused profile that helps teams empathize with users, uncover hidden needs, and design products or services that truly resonate with their audience.

Target Users

The primary users of this visualization tool in India include daily commuters in metropolitan cities like Delhi, Mumbai, and Bengaluru, EV scooter owners, ride-sharing drivers, delivery fleet operators, and first-time buyers transitioning from petrol vehicles to electric vehicles. These users are cost-conscious, practical, and highly influenced by infrastructure availability and fuel price trends.

Think & Feel

Indian EV users often feel anxious about whether their vehicle will have enough charge to complete daily travel, especially in cities with heavy traffic. They worry about limited charging infrastructure, battery degradation, and high battery replacement costs. Many users are also concerned about performance during extreme summer temperatures. At the same time, they feel motivated by rising petrol prices and the desire to save money while contributing to a cleaner environment.

See

Users observe increasing EV adoption on Indian roads, especially electric scooters and compact cars. They see government initiatives and subsidies promoting electric mobility. However, they also notice uneven charging infrastructure across regions, crowded urban roads affecting range efficiency, and limited charging options in apartment complexes. Petrol price fluctuations further influence their perception of EV affordability.

Hear

Indian consumers frequently hear mixed opinions about EVs. Some say charging infrastructure is still developing, while others claim EVs are suitable only for city use. Media coverage highlights new EV launches and government policies supporting electric mobility. Friends, family, and social media discussions also shape their expectations regarding range reliability and charging time.

Say & Do

Users often check their battery percentage before starting a trip and prefer charging their vehicles overnight at home. They use mobile applications to locate nearby charging stations and plan routes carefully, especially for longer journeys. Many compare the running cost per kilometer with petrol vehicles and monitor energy consumption to maximize efficiency. They may avoid long-distance highway travel unless charging points are confirmed along the route.

Pains

Major pain points include range anxiety, limited public charging stations in certain areas, long charging times, and reduced battery performance in high temperatures. Apartment residents face challenges installing personal chargers. Users also struggle with the lack of real-time data on charger availability and inconsistent charging speeds.

Gains

Indian EV users seek accurate and reliable range prediction adjusted for traffic and weather conditions. They want real-time updates on nearby charger availability, cost comparison insights in rupees per kilometer, and clear battery health visualization. Ultimately, they desire confidence, cost savings, and stress-free travel while using their electric vehicle.

3.Brainstorming

EV: Brainstorm & Idea Prioritization

Brainstorming in the EV project creates a collaborative and open environment where team members generate innovative solutions for improving electric vehicle charge efficiency and range optimization.

During this process, all ideas are welcomed — especially data-driven, technology-based, and sustainable mobility solutions. Quantity of ideas is encouraged first, followed by structured evaluation based on impact, feasibility, and innovation.

This template helps our team analyze EV challenges such as charging infrastructure gaps, battery performance, and range prediction using analytics tools and dashboard visualization.

By organizing, grouping, and prioritizing ideas, we ensure that the most impactful and practical solutions are selected for implementation in our EV analytics platform.

Step-1: Team Gathering, Collaboration and Select the Problem Statement

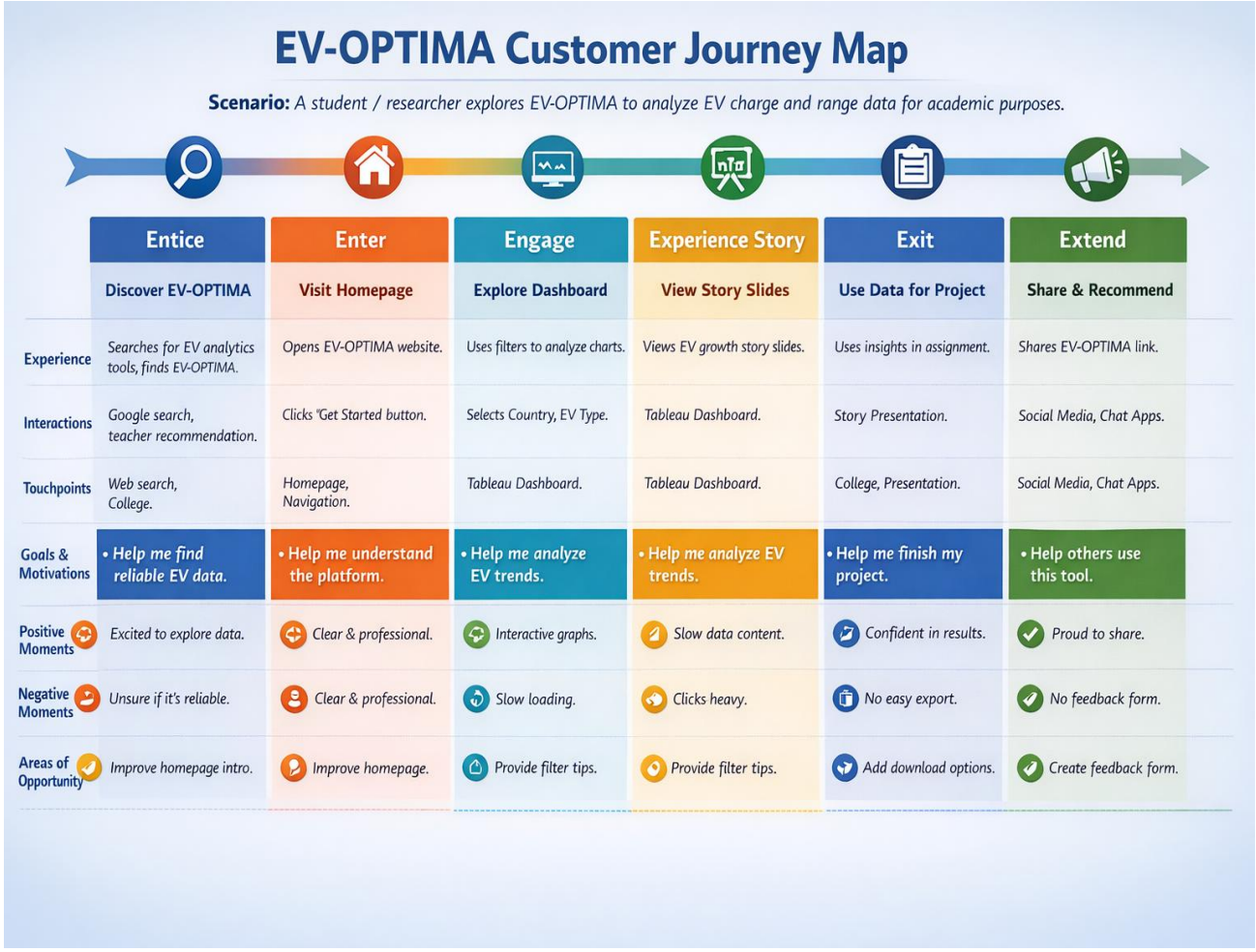
Step-2: Brainstorm, Idea Listing and Grouping

Step-3: Idea Prioritization

3.REQUIREMENT ANALYSIS

1. Customer Journey map

This diagram presents the **Customer Journey Map** for **EV-OPTIMA**, a tool used by students and researchers to analyze electric vehicle (EV) charge and range data for academic purposes.



It highlights six stages: discovering the tool, visiting the homepage, exploring dashboards, viewing EV growth stories, using data for projects, and sharing the tool. At each stage, users face positive experiences like clear design and interactive graphs but also challenges such as slow loading and limited export options. The map identifies opportunities to improve the homepage, optimize performance, add export features, and create a feedback form to enhance user satisfaction.

2.Solution Requirement

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	**Functional Requirement (Epic)**	**Sub Requirement (Story / Sub-Task)**
FR-1	Real-Time Range Estimation	Calculate remaining driving range based on current battery % Adjust range based on driving style, traffic, terrain, and weather Visualize predicted range on mobile and in-car dashboard
FR-2	Charging Station Map Integration	Show nearby chargers Display type & availability
FR-3	Trip Planning & Route Optimization	Suggest optimal charging stops along planned routes Calculate total trip time including estimated charging duration
FR-4	Battery Health Monitoring	Monitor battery degradation trends over time Send alerts for abnormal battery performance or maintenance recommendations

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
FR-1	Usability	The system should be simple, user-friendly, and easy to navigate
FR-2	Security	User data and login information should be securely protected.
FR-3	Reliability	The dashboard should provide accurate and consistent analysis results.
FR-4	Performance	The dashboard should load fast and respond quickly to user actions
NFR-5	Availability	High uptime (>99%); accurate charger and traffic data
NFR-6	Scalability	Handle growing users and data without performance loss

3.Data Flow Diagram

The EV Charge and Range Analysis System collects real-time data from the vehicle's sensors, charging stations, and external sources like traffic and weather services. It processes this data to estimate the remaining driving range, suggest optimal charging stations, and provide trip planning. Users interact through a mobile or web app where they input preferences and receive notifications, alerts, and visualizations about battery status, charging availability, and route guidance.

EV Charge and Range Analysis Data Flow Diagram

1.User (EV Driver / Researcher)

The primary actor interacting with the system.

Vehicle preferences, trip plans, route choices, personal settings.

Receives battery status, charging station suggestions, route guidance, and alerts.

To plan trips efficiently, monitor EV range, and reduce range anxiety.

2.EV Onboard System / Vehicle Sensors

Internal sensors and onboard diagnostics of the electric vehicle.

None directly; it continuously collects vehicle data.

Sends battery percentage, speed, location, and energy consumption data to the system.

Provides real-time vehicle status for accurate range prediction and analysis.

3.Charging Station Network

External infrastructure providing EV charging services.

User queries for nearby charging stations.

Sends charger availability, type (fast/slow), location, and operational status.

4. Traffic and Weather Data Services

External data sources providing environmental conditions.

System requests for relevant route conditions.

Real-time traffic updates, road conditions, and weather data.

To improve range estimation and suggest optimized routes by considering traffic delays or adverse weather.

5. **EV Charge & Range Analysis System**

Central processing system of the application.

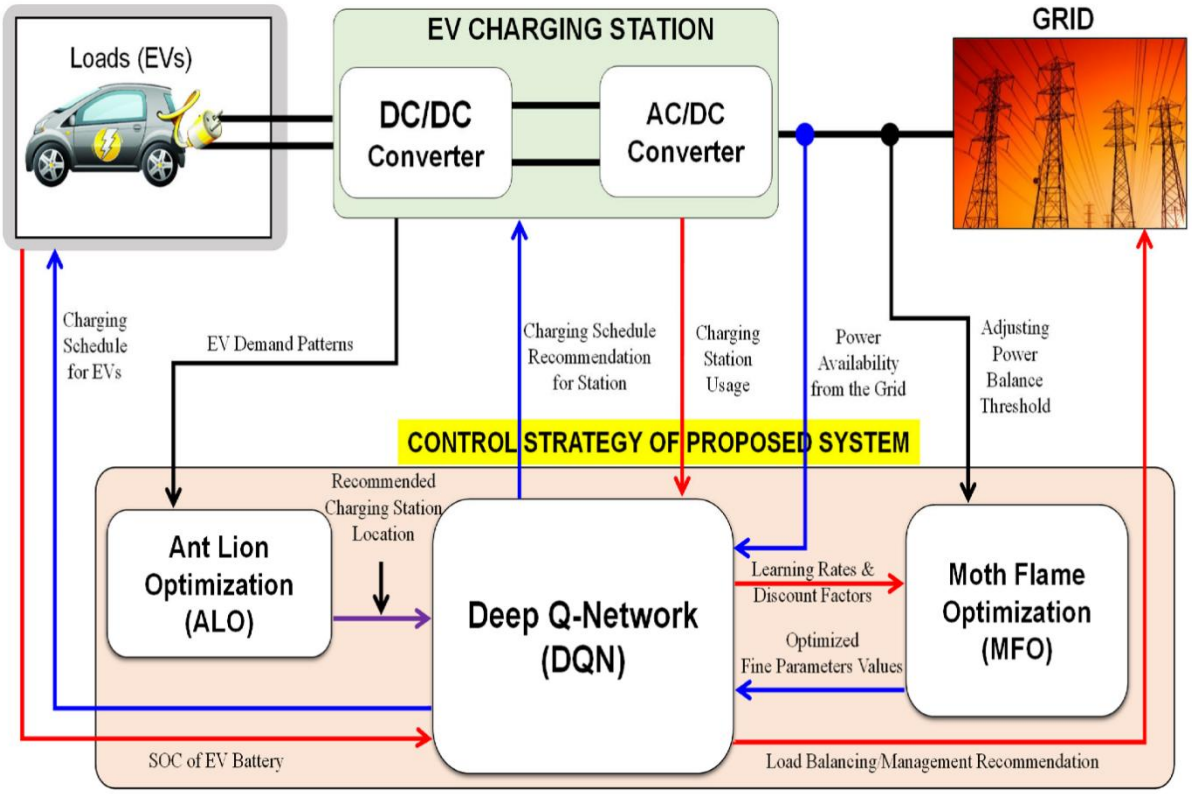
Data from EV sensors, charging stations, traffic/weather services, and user preferences.

6.User Interface (Mobile App / Web Dashboard)

Interface for interaction between the user and the system.

Receives user inputs like preferences, trip plans, and filters.

Displays battery status, range prediction, charging station locations, route guidance, and alerts.



4.Technology Stack

This diagram shows the flow of data in the system. The User interacts with the Web UI (Tableau Dashboard). The request is processed in the Application Logic (Python/Tableau), which accesses data from the Database. Finally, the system displays Visualization & Insights to help users understand and analyze the data easily.

4.PROJECT DESIGN

1.Problem Solution Fit

In our project, **"Visualization Tool for Electric Vehicle Charge and Range Analysis"**, we observed that EV users and fleet operators often face difficulty in accurately monitoring battery charge levels, predicting driving range, and analyzing energy consumption patterns. Inconsistent range estimation, inefficient charging management, and lack of real-time insights can lead to range anxiety, poor trip planning, and reduced vehicle performance.

These challenges are commonly caused by improper data monitoring, lack of predictive analysis, inefficient battery usage tracking, and absence of structured visualization systems. As a result, EV performance analysis becomes complex, less reliable, and time-consuming, which can impact decision-making for both individual users and organizations.

To overcome these challenges, our project focuses on developing an interactive visualization system for effective EV charge and range analysis, including:

Battery Performance Monitoring:

Tracking real-time battery charge percentage, voltage, current, and temperature.

Analyzing charge–discharge cycles to evaluate battery health and efficiency.

Range Prediction & Analysis:

Developing predictive models to estimate remaining driving range based on speed, terrain, and driving behavior.

Visualizing energy consumption per kilometer for better trip planning.

Data Optimization & Processing:

Filtering and cleaning sensor data to ensure accurate range estimation.

Pre-processing large EV datasets to improve dashboard performance and responsiveness.

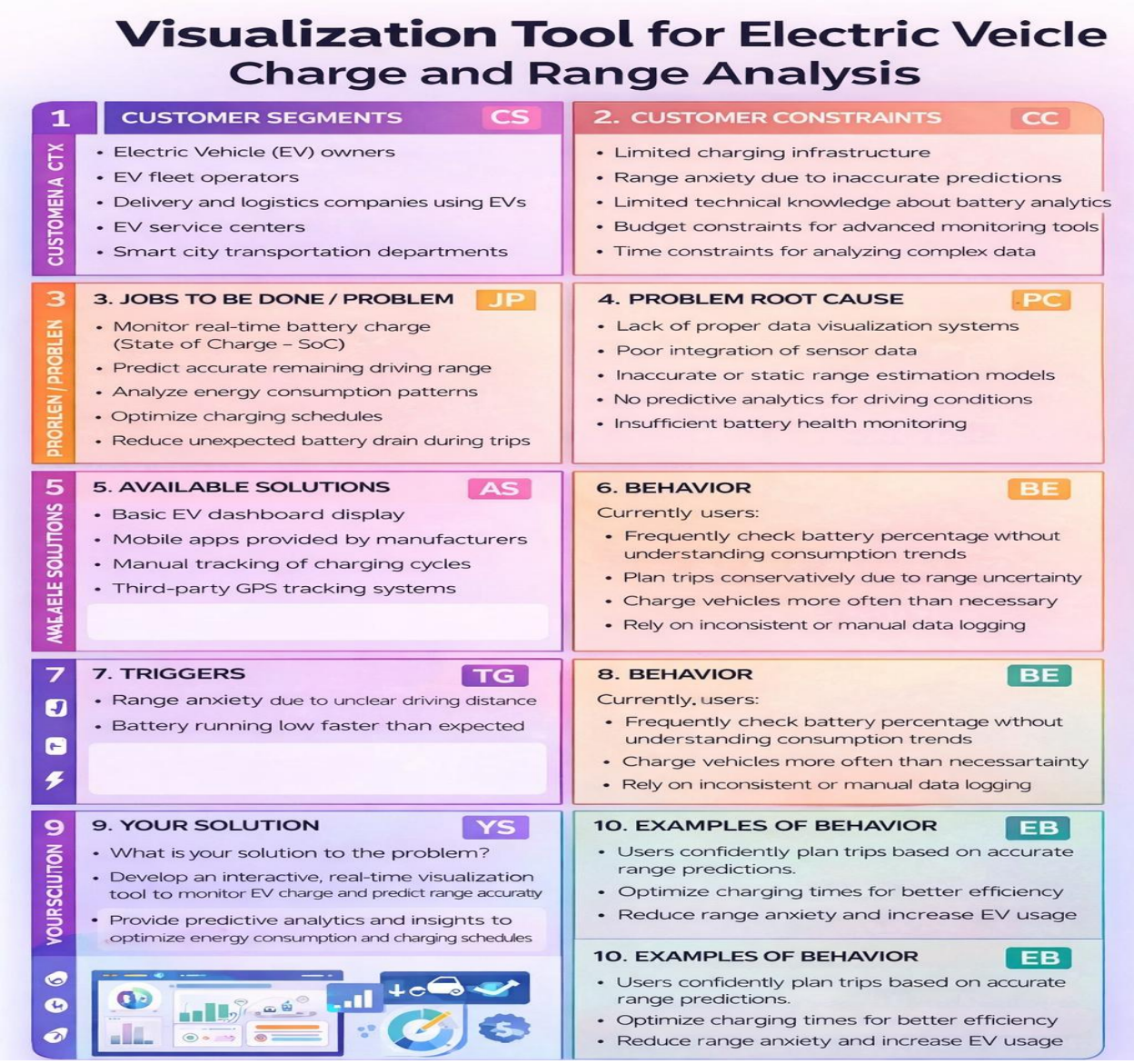
Dashboard Design Optimization:

Designing interactive dashboards to display:

- State of Charge (SoC)
- Estimated Remaining Range

- Energy Consumption Trends
- Charging Time Analysis

Using efficient chart types and minimal visual clutter to ensure fast rendering and user-friendly analysis.



2.Proposed solution Details:

S.No	Parameter	Description
1	State of Charge (SoC)	Indicates the current battery charge level in percentage.
2	State of Health (SoH) .	Represents the overall condition and aging level of the battery.
3	Battery Voltage	Measures the real-time voltage supplied by the battery.
4	Battery Current	Shows the current flow during charging and discharging.
5	Battery Temperature	Monitors temperature to prevent overheating and ensure safety.
6	Remaining Range	Estimates the distance the vehicle can travel with the available charge.

3. Solution Architecture

The **Visualization Tool for Electric Vehicle Charge and Range Analysis** project analyzes EV battery performance and driving range using real-time and historical data. The architecture illustrates how data flows from the electric vehicle sensors to meaningful visual insights for users.

Flow:

EV Sensors → Data Collection Module → Application Logic (Python/Analytics Engine) → Database → Web UI (Dashboard) → Visualization & Insights

EV Sensors:

Collect real-time data such as battery charge level (SoC), voltage, current, temperature, speed, and energy consumption.

Data Collection Module:

Captures and transfers sensor data to the processing system in structured format.

Application Logic:

Processes raw data, performs range prediction calculations, analyzes energy consumption patterns, and generates insights.

Database:

Stores EV data including battery parameters, trip history, charging sessions, and performance metrics.

Web UI (Dashboard):

Provides an interactive platform where users can monitor battery status, analyze range predictions, and review energy trends.

Visualization & Insights:

Displays charts, graphs, alerts, and comparative analysis to support better trip planning and charging decisions.

Goals:

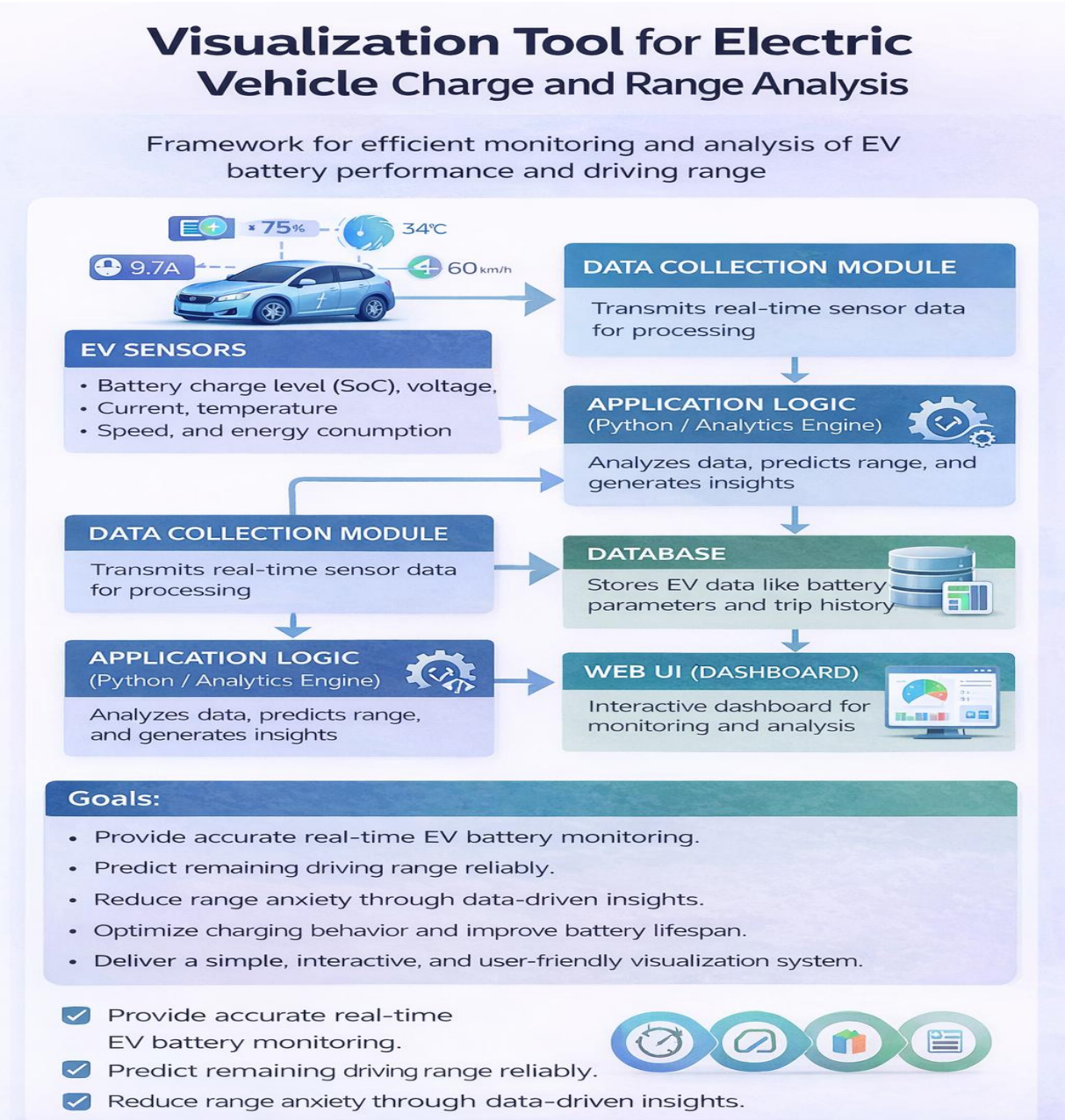
Provide accurate real-time monitoring of EV battery performance.

Predict remaining driving range reliably.

Reduce range anxiety through data-driven insights.

Optimize charging behavior and improve battery lifespan.

Solution Architecture Diagram:



5.PROJECT PLANNING & SCHEDULING

Project Planning

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint -1	Data Collection	USN-1	Data Collection & Extraction from Database	3	High	M.Sasikala
Sprint -1	Data Preparation	USN-2	Data cleaning and preprocessing	2	High	M.Sasikala
Sprint -2	Data Visualization	USN-3	Create charts and graphs using Tableau	3	High	M.Dhilleswari , M.Ashajyothi
Sprint -2	Dashboard	USN-4	Develop interactive dashboard	3	High	M.Dhilleswari
Sprint -3	Story	USN-5	Create Tableau Story for analysis	2	Medium	M.Sasikala

Sprint -4	Publishing and Web Integration	USN-6	Publish dashboard to web	2	Mediu m	M.Ashajyothi
Sprint -5	Publishing and Web Integration	USN-7	Final demo & documentati on submission	2	High	M.Sasikala M.Ashajyothi M.Dhilleswari

6.FUNCTIONAL AND PERFORMANCE TESTING

Performance Testing

Model Performance Testing:

Model Performance Testing evaluates the accuracy, reliability, and efficiency of the predictive models used in the system—especially the **range prediction model** and **energy consumption analysis model**. Since the project includes analytics to estimate the remaining driving range and battery behavior, it is important to test how well these models perform under different real-world conditions.

The range prediction model is tested using historical trip data, varying speed patterns, different battery charge levels, and environmental factors. The predicted range is compared with the actual distance traveled to measure accuracy. Metrics such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and prediction accuracy percentage are used to evaluate performance.

Additionally, the model's response time is tested to ensure predictions are generated quickly without causing dashboard delays. Stability testing is also performed to verify consistent predictions across multiple datasets. These tests confirm that the predictive model provides reliable insights and supports effective trip planning and charging decisions.

2.Proposed Solution

S.No	Parameter	Screenshot / Values
1	Data Rendered	Dataset contains: EV Model, Battery Capacity (kWh), Range (km), Charging Time (hrs), Charging Type, Price, Region. Total Records: 500+. Data successfully loaded into Tableau Data Pane
2	Data Preprocessing	Removed duplicates (5 records). Handled missing values in Charging Time. Converted Range & Battery Capacity into numeric format. Created hierarchy: Region → City.
3	Utilization of Filters	Filters Used: EV Model, Charging Type (Fast/Slow), Battery Capacity Range, Region. Screenshot shows interactive filter panel applied to dashboard
4	Calculated Fields Used	Efficiency = Range / Battery Capacity 2. Charging Cost per 100km = (Electricity Cost × Battery Capacity) / Range 3. Average Range = AVG(Range)
5	Dashboard Design	No of Visualizations / Graphs – 6 1. Range by EV Model (Bar Chart) 2. Charging Time vs Battery Capacity (Scatter Plot) 3. Range Trend Over Years (Line Chart) 4. Charging Type Comparison (Bar Chart) 5. Region-wise EV Distribution (Map) 6. KPI Cards (Avg Range, Avg Charging Time)
6	Story Design	No of Visualizations / Graphs – 5 Story Points 1. EV Market Overview 2. Range Comparison 3. Charging Performance Analysis 4. Regional Insights 5. Strategic Recommendations

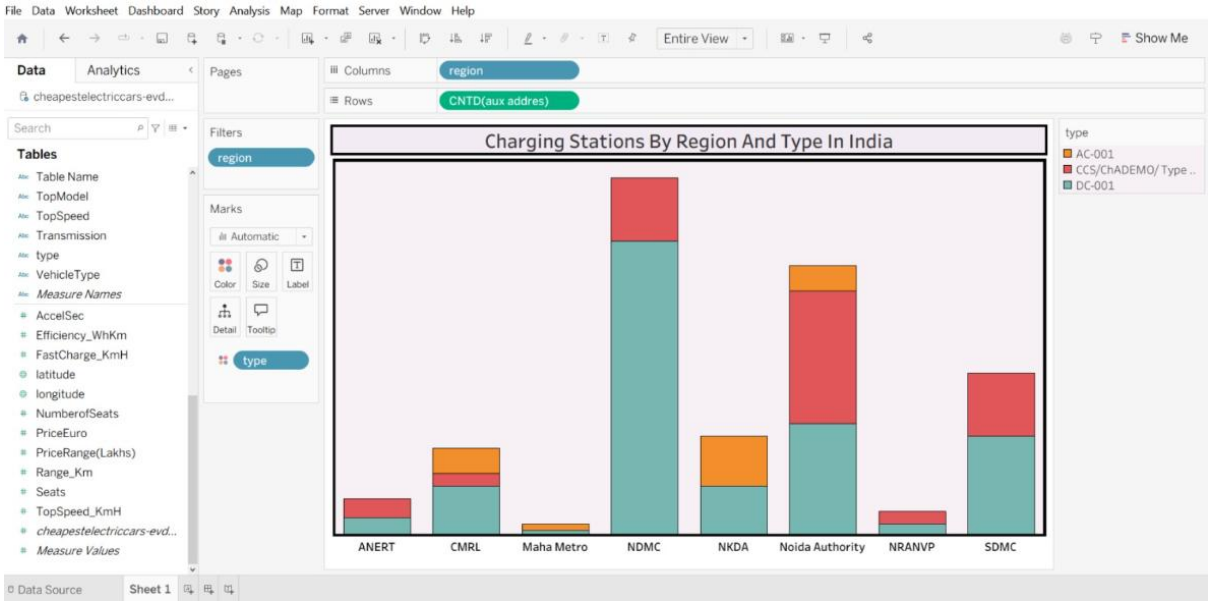
7.RESULTS

Output Screenshots

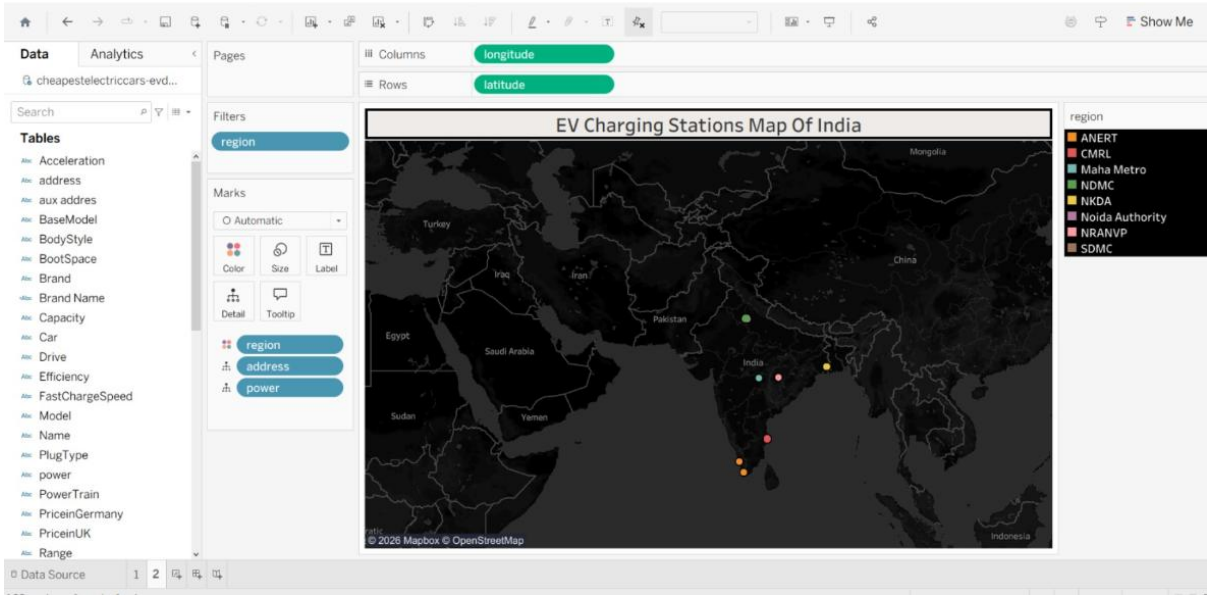
Visualizations

We created several visualizations:

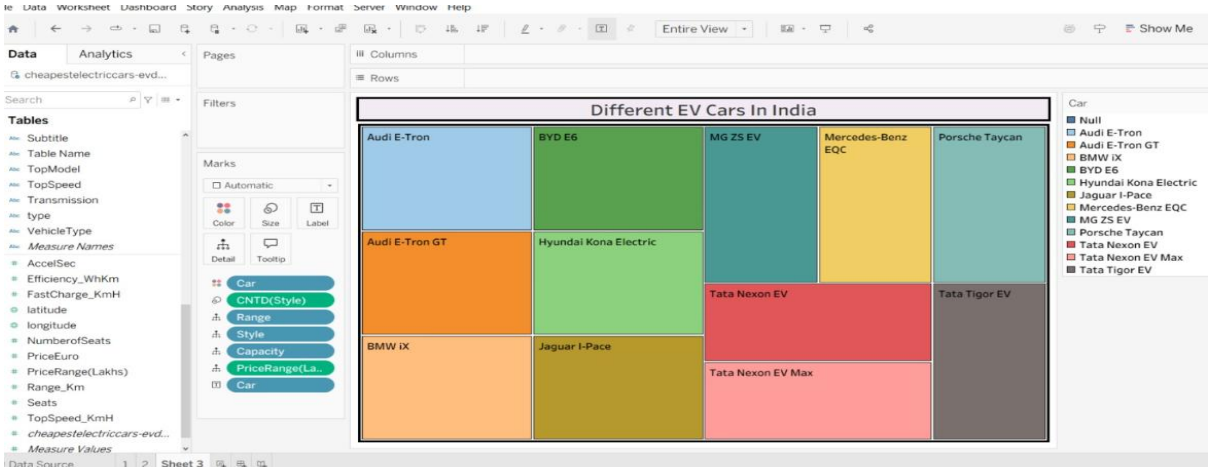
1.Charging stations By And Type In India



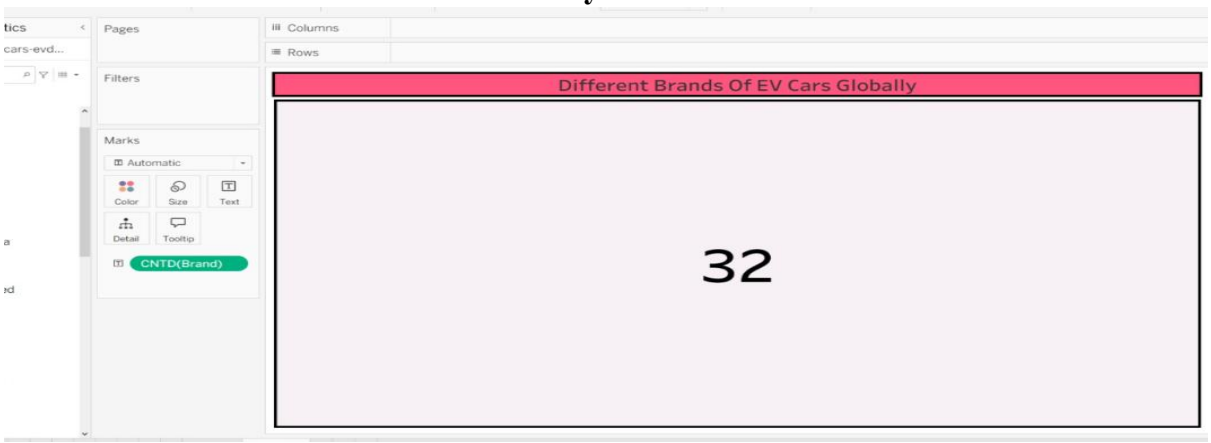
2.EV Charging Stations Map Of India



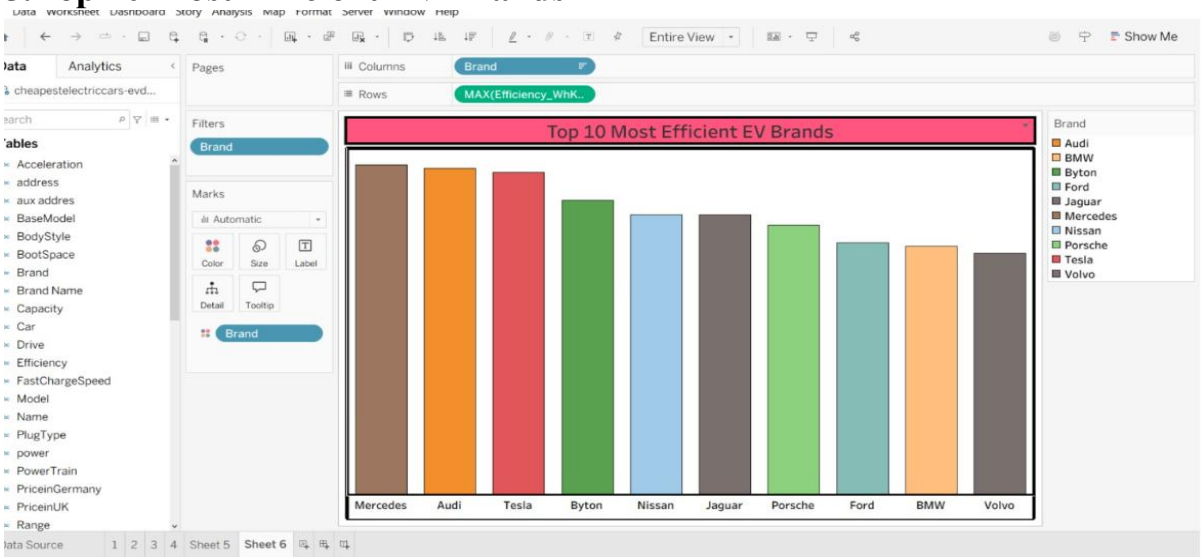
3.Different EV Cars In India



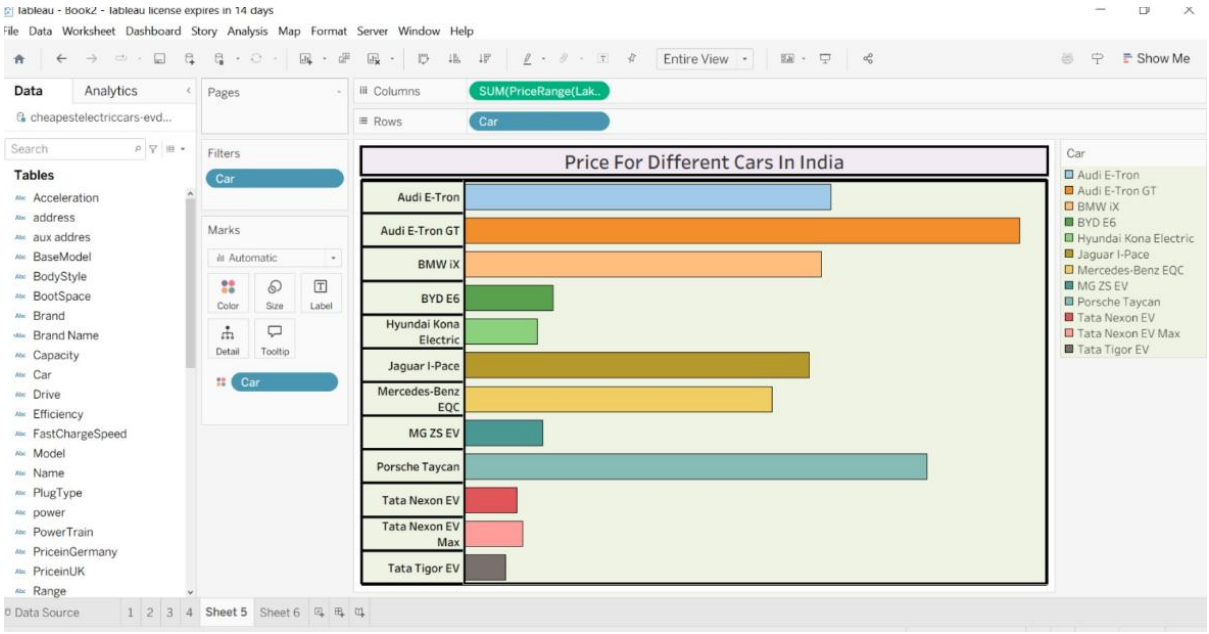
4.Different Brands Of EV Cars Globally



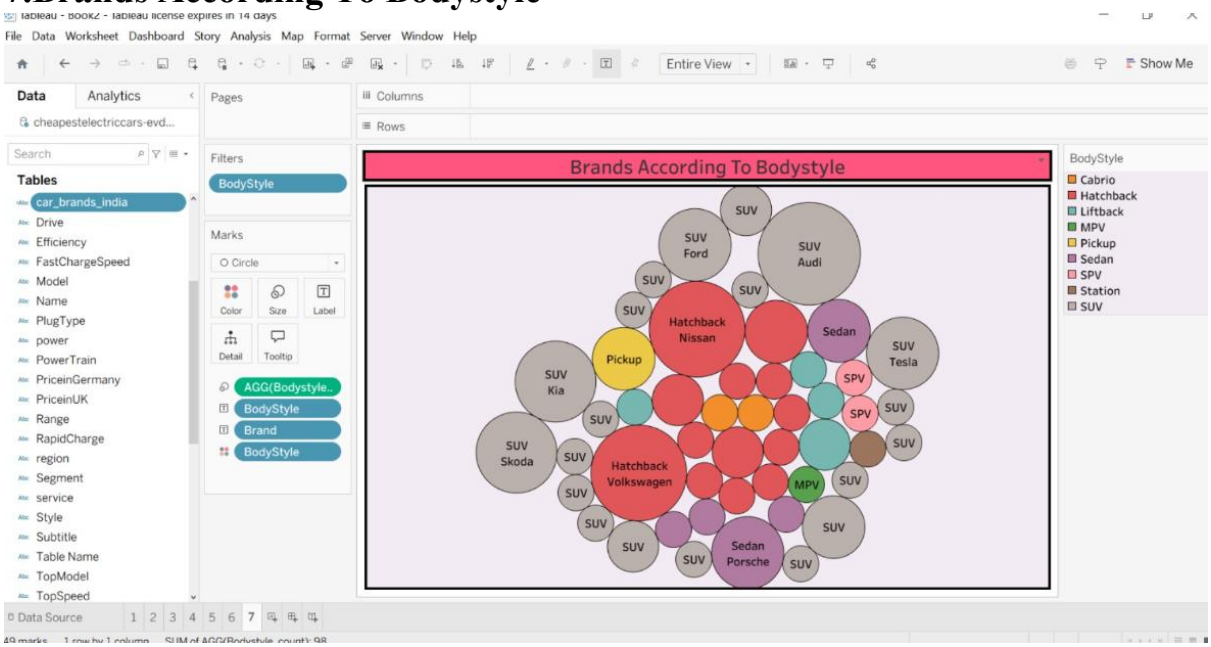
5.Top 10 Most Efficient EV Brands



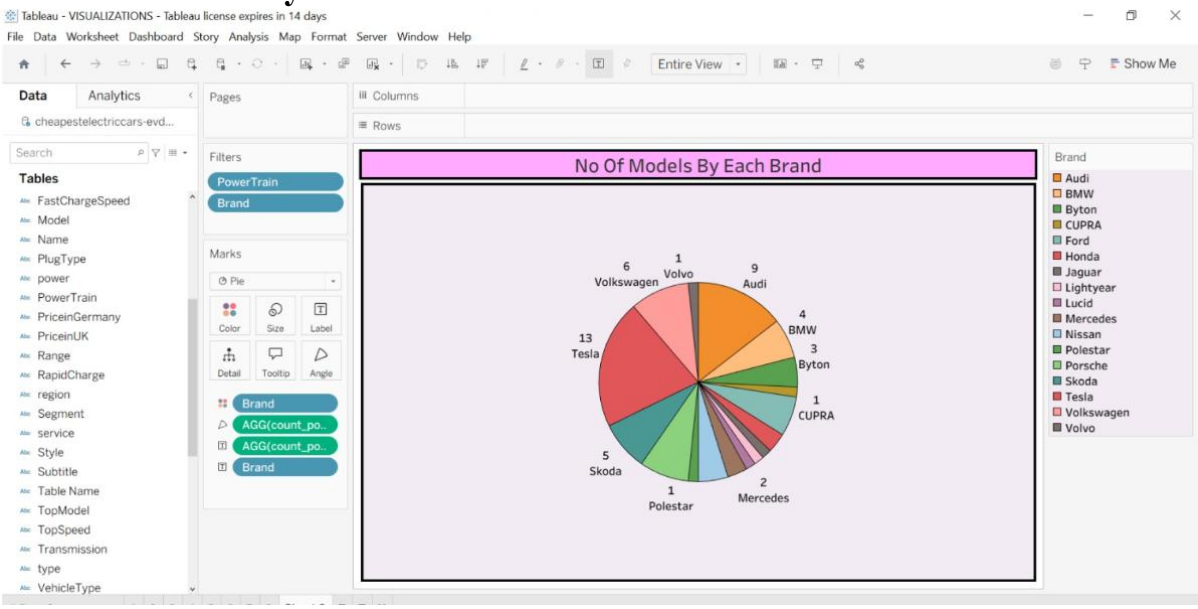
6.Price For Different Cars In India



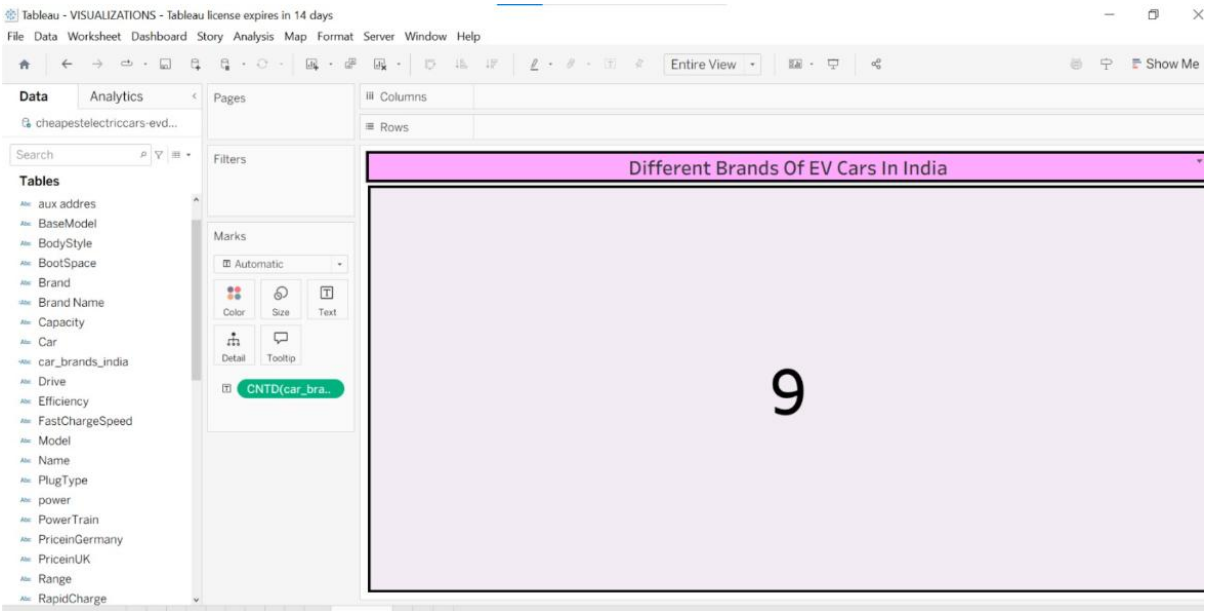
7.Brands According To Bodystyle



8.No Of Models By Each Brand



9.Different Brands Of Cars In India



Dashboard:

Dashboard Name:**Visualization Tool for Electric Vehicle Charge and Range Analysis**

Dashboard Description Project

Title:

EVision – A Data-Driven Exploration of the Global and Indian Electric Vehicle Market

The dashboard is designed to provide interactive and data-driven insights into the Electric Vehicle (EV) ecosystem across the globe and in India. It highlights brand distribution, efficiency comparison, body style segmentation, model availability, and performance metrics such as top speed. The dashboard enables users to analyze market presence, competitive positioning, and performance trends in the EV industry.

Dashboard Components:

KPI Cards – Total EV Brands (Global & India)
Displays the total number of EV brands worldwide and the number of brands available in India.

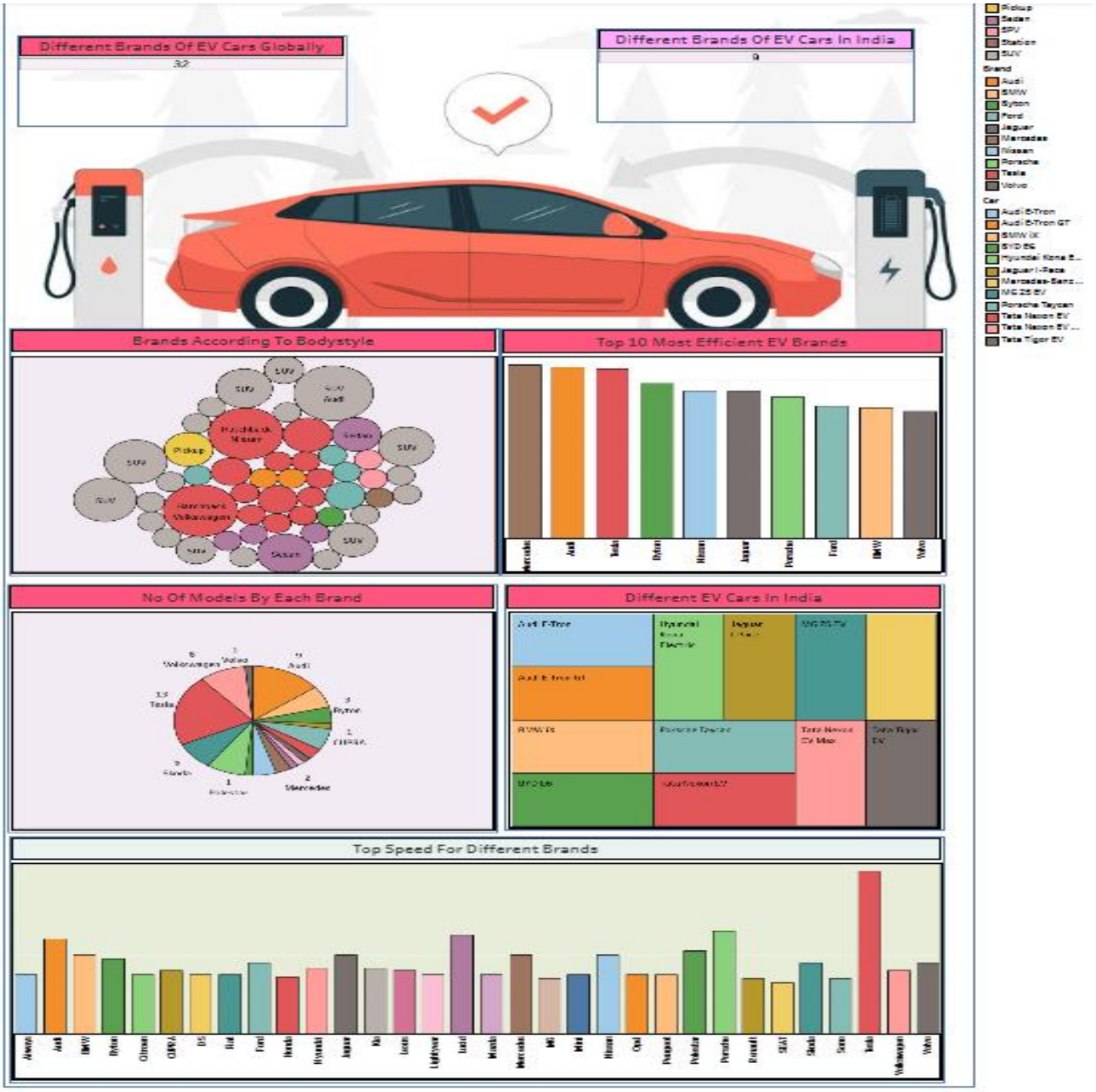
Bubble Chart – Brands According to Bodystyle
Visualizes the distribution of EV brands across body types such as SUV, Sedan, Hatchback, Pickup, and Coupe.

Bar Chart – Top 10 Most Efficient EV Brands
Ranks EV brands based on energy efficiency performance.

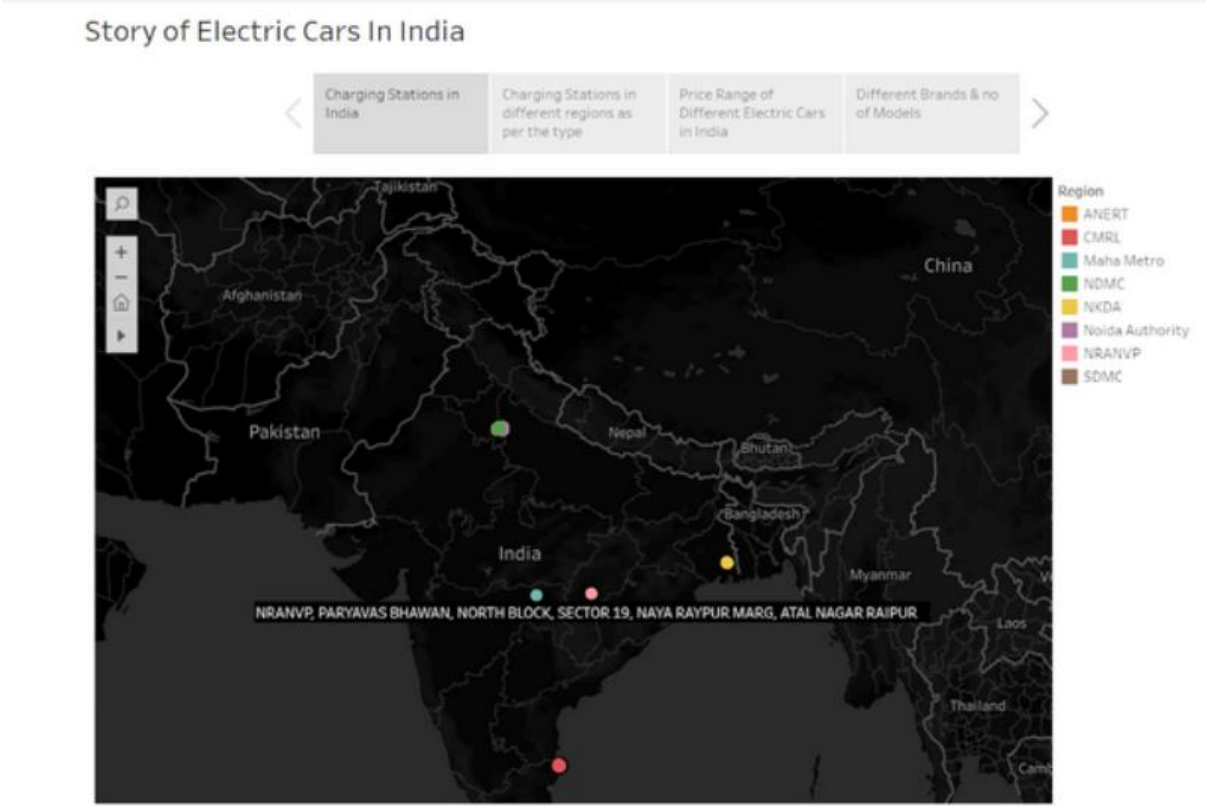
Pie Chart – Number of Models by Each Brand
Shows brand-wise distribution of available EV models.

Treemap – Different EV Cars in India
Displays EV models available in India with proportional representation.

Bar Chart – Top Speed for Different Brands
Compares maximum speed performance across EV brands.


Different Brands Of EV Cars Globally
32
Different Brands Of EV Cars In India
9
Brands According To Bodystyle
Top 10 Most Efficient EV Brands
No Of Models By Each Brand
Different EV Cars In India
Top Speed For Different Brands
Pickup
Sedan
SPV
Station
SUV
Brand
Audi
BMW
Byton
Ford
Jaguar
Mercedes
Nissan
Porsche
Tesla
Volvo
Car
Audi E-Tron
Audi E-Tron GT
BMW iX
BYD E6
Hyundai Kona E...
Jaguar I-Pace
Mercedes-Benz ...
MG ZS EV
Porsche Taycan
Tata Nexon EV
Tata Nexon EV ...
Tata Tigor EV

Story



8.ADVANTAGES & DISADVANTAGES

Advantages of a Visualization Tool for Electric Vehicle Charge and Range Analysis:

Improved Understanding of Battery Performance

A visualization tool presents complex battery data—such as state of charge (SoC), energy consumption, and remaining range—in a clear and graphical format. Instead of interpreting raw numerical values, users can observe patterns and trends over time. This makes it easier to understand how driving behavior and environmental conditions affect battery usage.

Enhanced Range Prediction Accuracy

By integrating real-time and historical data such as speed, terrain, weather conditions, and HVAC usage, visualization tools provide more realistic range estimates. This helps drivers make informed decisions and reduces uncertainty compared to relying solely on manufacturer-stated range figures.

Better Trip Planning and Route Optimization

Visualization platforms often include route-mapping features that identify optimal charging stops and estimate charging time required during a trip. By displaying projected battery levels at different points along the route, these tools help reduce range anxiety and improve travel efficiency.

Cost and Energy Efficiency Monitoring

Such tools allow users to track electricity consumption, cost per kWh, and energy usage per kilometer or mile. Over time, this enables better budgeting and helps identify opportunities to improve energy efficiency, especially for commercial EV fleets.

Battery Health and Sustainability Insights

Long-term visualization of battery degradation supports maintenance planning and warranty tracking. Additionally, some tools estimate carbon emission savings, contributing to sustainability analysis and reporting.

Disadvantages of a Visualization Tool for Electric Vehicle Charge and Range Analysis:

Dependence on Data Accuracy

The reliability of insights generated by visualization tools depends heavily on the accuracy of input data. Incorrect or incomplete data—such as outdated battery health metrics or inaccurate environmental inputs—can lead to misleading range predictions.

Risk of Oversimplification

Some visualization tools rely on simplified assumptions about driving conditions. Real-world factors like traffic congestion, aggressive driving, or sudden weather changes may not be fully captured, which can reduce predictive accuracy.

Privacy and Data Security Concerns

These tools often collect sensitive data, including location history, charging behavior, and driving patterns. Without proper cybersecurity measures, this information could be vulnerable to misuse or unauthorized access.

Integration and Compatibility Challenges

Different electric vehicle manufacturers use proprietary data systems and APIs. This can create compatibility issues, making it difficult for visualization tools to integrate seamlessly across multiple EV brands.

High Development and Maintenance Costs

Building and maintaining an effective visualization platform requires continuous software updates, cloud infrastructure, and data analytics capabilities. For startups or research institutions, these requirements may result in significant operational costs.

Conclusion

A Visualization Tool for Electric Vehicle Charge and Range Analysis plays a crucial role in enhancing the efficiency, reliability, and user experience of electric vehicle operations. By transforming complex battery and energy data into clear graphical insights, it enables drivers, fleet managers, and researchers to better understand charge levels, predict range, monitor battery health, and optimize trip planning. These tools significantly reduce range anxiety and support data-driven decision-making, ultimately contributing to improved energy efficiency and cost management.

However, the effectiveness of such tools depends largely on data accuracy, system integration, and cybersecurity measures. Inaccurate inputs, oversimplified modeling, or privacy vulnerabilities can limit their reliability and user trust. Additionally, development and maintenance costs may present challenges for widespread implementation.

Overall, despite certain limitations, visualization tools represent a valuable technological advancement in the electric vehicle ecosystem. With continuous improvements in data analytics, connectivity, and machine learning models, these tools are expected to become increasingly precise, secure, and essential for sustainable transportation management.

10.FUTURE SCOPE

Integration with Artificial Intelligence and Machine Learning

Future developments can incorporate advanced AI and machine learning algorithms to improve range prediction accuracy. By analyzing historical driving behavior, terrain, weather patterns, and battery degradation trends, the system can generate highly personalized and adaptive range forecasts.

Real-Time IoT and Smart Grid Connectivity

Integration with Internet of Things (IoT) devices and smart grid systems can enable real-time energy management. The tool could optimize charging schedules based on electricity tariffs, grid load, and renewable energy availability, promoting cost savings and sustainable energy use.

Advanced Battery Health Analytics

Future versions may include predictive maintenance features that detect early signs of battery degradation or faults. Using big data analytics, the tool could forecast battery lifespan and recommend optimal maintenance schedules.

Vehicle-to-Grid (V2G) Integration

With the growth of Vehicle-to-Grid technology, the visualization tool can support bidirectional energy flow analysis. It could help users monitor energy supplied back to the grid and calculate financial incentives or energy credits earned.

Augmented Reality (AR) and Enhanced User Interfaces

Advanced user interfaces, including augmented reality dashboards or voice-assisted systems, could provide more interactive and intuitive data visualization, improving accessibility for a broader range of users.

Multi-Vehicle Fleet Optimization

For commercial applications, future systems could integrate fleet-wide analytics to optimize route planning, charging infrastructure placement, and overall energy efficiency across multiple vehicles.

Integration with Autonomous and Connected Vehicles

As autonomous EV technology evolves, visualization tools can be integrated with autonomous driving systems to dynamically adjust routes and charging plans without human intervention.

Blockchain for Secure Data Management

Blockchain technology could enhance data security, ensuring transparent and tamper-proof records of charging history, battery health, and energy transactions.

11.APPENDIX

Source Code

- app.py — Flask application
- data_loader.py — Data pipeline
- templates/index.html — Frontend
- static/css/style.css — Styling
- static/js/main.js — JavaScript

Dataset Link

https://drive.google.com/drive/folders/1Rkzdks6Us1Uq2SRB4nxMAb83jN5bpHll?usp=drive_link

Tableau public link

https://public.tableau.com/app/profile/sasikala.mannena/viz/StoryElectriccarsinIndia_17722097711430/Dashboard1?publish=yes

GitHub link:

https://github.com/Sasikala181/visualization-tool-for-electric-vehicle-charge-and-range-analysis

Project Demo Link

The project demonstration video showing the implementation and working process of the Visualization Tool for Electric Vehicle Charge and Range Analysis dashboard can be accessed using the below link:

https://drive.google.com/file/d/1mAkI0Ikh1aFxICFssseiObtbz-_VYngU/view?usp=drivesdk